
Behavioural Evaluation of Post-Training Quantisation Across Transfer Learning Regimes in TinyML

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Abstract

Tiny Machine Learning (TinyML) enables deep neural networks to run on resource-constrained edge devices, where post-training quantisation (PTQ) and transfer learning are commonly used to address hardware and data constraints. However, PTQ in TinyML is typically evaluated under a fixed training regime using aggregate accuracy and FP32 to INT8 degradation, potentially overlooking behavioural differences across transfer learning regimes. This study examines how four transfer learning regimes interact with PTQ for MobileNetV2-0.35 on PlantVillage. Beyond accuracy, behaviour evaluation includes KL divergence and prediction flip rates between FP32 and INT8 models, class-level differences and robustness to Gaussian noise and JPEG corruption. Regimes allowing backbone adaptation achieve high INT8 accuracy with relatively small degradation, whereas only the backbone-frozen regime shows lower INT8 performance and greater degradation. Across the stronger regimes, behavioural patterns remain broadly aligned, with some differences emerging under severe corruption. Overall, aggregate metrics appear to capture most practical post-quantisation differences across regimes.

1. Introduction

Machine learning inference has traditionally been performed in the cloud, but this can introduce latency, transmission energy costs, and privacy concerns when sensitive data must be sent to remote servers. These limitations have motivated a shift towards edge-based machine learning. TinyML

focuses on deploying deep learning models on microcontroller-class devices with extremely limited memory and computational resources. To enable deployment in such constrained environments, quantisation reduces memory usage and computational cost by lowering the numerical precision used for model parameters and arithmetic operations (Tsoukas et al., 2024). In practice, TinyML systems often rely on post-training quantisation (PTQ), which compresses models without requiring additional retraining and integrates naturally with existing microcontroller deployment toolchains. However, PTQ can introduce performance degradation (Nagel et al., 2021).

Beyond hardware limitations, TinyML deployments are often constrained by limited task-specific labelled data, as models must be adapted to highly specialised tasks and environments where training from scratch is impractical. Transfer learning addresses this constraint by adapting pretrained models using relatively small training datasets (Hayajneh et al., 2024). Specifically, commonly used transfer learning strategies such as Linear Probing and Full Fine-Tuning differ in the extent to which network parameters are updated and adapted to the target dataset, which may result in changes to weight and activation distributions. These distributional differences may influence PTQ behaviour and quantisation sensitivity (Nagel et al., 2021).

However, PTQ is commonly evaluated under a fixed training regime, so the influence of transfer learning

strategy on quantisation behaviour and sensitivity remains less well understood. Moreover, PTQ is typically assessed using aggregate INT8 accuracy and FP32 to INT8 degradation; such metrics may not fully capture post-quantisation behavioural differences across regimes. This study therefore asks whether PTQ degradation generally differs across transfer learning regimes, and whether regimes achieving strong INT8 accuracy also show broadly similar post-quantisation behaviour. To address this, we evaluate four transfer learning regimes under a unified configuration and examine FP32 to INT8 degradation, output-level behavioural metrics comparing each FP32 model with its INT8 counterpart, and robustness to input corruption.

2. Literature Review

2.1 TinyML

Conventional ML deployments on embedded and edge devices often stream raw data to the cloud for inference, introducing latency and raising privacy concerns. TinyML instead deploys models directly on microcontroller-class devices, enabling low-latency, privacy-preserving operation with reduced reliance on cloud connectivity. Typical TinyML platforms provide around 256-512 kB SRAM and 1-2MB flash (Brockman et al., 2024), making conventional deep neural networks impractical and motivating compact architectures together with aggressive model compression techniques (Tsoukas et al., 2024). Among these compression strategies, post-training quantisation is widely adopted due to its simplicity and compatibility with pretrained models (Nagel et al., 2021), motivating analysis of how training strategies and model adaptation interact with quantisation.

2.2 Post-Training Quantisation

Post-training Quantisation (PTQ) is a model compression technique applied after training that reduces the numerical precision of network parameters and intermediate activations, typically converting 32-bit floating-point values to 8-bit integers. This reduction lowers memory requirements and enables inference to be executed using integer arithmetic, which is generally more efficient on resource constrained hardware. This makes PTQ well suited to edge and embedded deployment settings where memory and

compute budgets are limited. However, reducing numerical precision introduces approximation error, which may degrade predictive performance.

Performance degradation in PTQ arises from quantisation error when floating-point values of weights and activations are mapped to a limited integer grid. The chosen floating-point range determines the quantisation scale: if the range is too narrow, values may be clipped, whereas if it is too wide, rounding error increases. For activations, per-tensor quantisation is typically used, with ranges estimated from calibration data using observer modules, so unrepresentative data or outliers can lead to suboptimal scaling and reduced accuracy (Nagel et al., 2021). For weights, using a single scale across an entire layer can disproportionately affect channels with smaller magnitudes, so per-channel quantisation is commonly used (Krishnamoorthi, 2018). As a result, the quantisation range, observer choice, and quantisation granularity influence how quantisation error is distributed and can affect overall model accuracy.

2.3 Transfer Learning

Transfer Learning (TL) enables models pretrained on large-scale datasets, such as ImageNet, to reuse learned feature representations for new tasks, thereby reducing the need for large labelled datasets and training from scratch. This approach is particularly well suited to TinyML settings, where labelled data are often limited and efficient adaptation for resource-constrained edge hardware is required (Hayajneh et al., 2024).

In CNN based literature, transfer learning is commonly implemented through different degrees of parameter adaptation. Linear Probing freezes the pretrained backbone and trains only a lightweight classifier head, offering low training cost but limited adaptability. Full Fine-Tuning updates all backbone parameters, enabling stronger task specific adaptation at higher computational cost. Between these extremes, parameter-efficient fine-tuning methods update only a small subset of parameters. One such method is Batch Normalisation Tuning, in which only the batch normalisation affine parameters, scale γ and shift β , are updated while convolutional weights remain fixed (Kanavati & Tsuneki, 2021). Knowledge Distillation instead modifies the training objective by encouraging a student model to match

the softened outputs of a teacher network alongside the ground truth labels (Hinton et al., 2015) and can be combined with either full or partial fine-tuning. Further details of Knowledge distillation are provided in Appendix A.1, with batch normalisation theory and folding discussed in Appendix A.2.

These regimes differ in how they alter weights and internal feature representations. Freezing most parameters preserves pretrained features, whereas broader adaptation can change the feature extractor and resulting activations (Kumar et al., 2022). Since post-training quantisation relies on weight and activation distributions to determine scaling parameters (Nagel et al., 2021), such differences may influence quantisation sensitivity across regimes. However, PTQ is typically evaluated using INT8 accuracy and FP32 to INT8 degradation under a fixed training regime (Lin et al., 2023), which may overlook behavioural stability after quantisation (Dutta et al., 2024). This study therefore extends prior work to a TinyML-oriented PTQ setting and examines whether degradation and behavioural patterns differ across regimes.

3. Methodology

3.1 Model Selection

Given the memory and storage constraints commonly associated with TinyML platforms, we select MobileNetV2-0.35 (Sandler et al., 2018) at 96x96 input resolution. This reduced-width variant uses fewer channels than standard MobileNetV2 and is widely adopted in TinyML vision studies (Lin et al., 2022; Brockman et al., 2024). For MobileNetV2-0.35, Lin et al. (2022) demonstrate feasibility within a 256 kB SRAM budget after INT8 quantisation, while Yang et al. (2025) report approximately 335 kB peak RAM usage during inference at the same resolution and observe sharp accuracy degradation for smaller width multipliers. Our MobileNetV2-0.35 model contains approximately 444,000 parameters, corresponding to an INT8 parameter storage of approximately 401 kB. Actual peak SRAM usage during inference will vary with device and implementation, so a fixed on-device memory footprint is not reported.

3.2 Dataset

We use the 38-class PlantVillage dataset (Hughes & Salathé, 2015), a widely used dataset for plant-disease recognition under controlled acquisition conditions. From each class, 100 non-augmented images are uniformly sampled (3,800 total) and split into 70/20/10 train, validation, and test sets, reflecting a relatively low-label regime. While the dataset consists of clean, studio-like images, this controlled setting isolates the effects of transfer learning and post-training quantisation without confounding factors such as background clutter or illumination variability. All images are resized to 96×96 pixels and normalised using the standard ImageNet mean and standard deviation. Training data is augmented using random horizontal flips, rotations ($\pm 15^\circ$), and colour jitter ($\pm 10\%$).

3.3 Transfer Learning Regimes

The transfer learning setup builds on the MobileNetV2 Linear Probing baseline of Seol et al. (2025), which was reproduced on the Li and Zhao (2019) concrete dataset to validate the pipeline. Models are trained using SGDM (momentum 0.9, batch size 50) for up to 200 epochs with early stopping (patience 10) and a ReduceLROnPlateau scheduler (patience 5). To reflect microcontroller-class constraints, TinyML experiments use ImageNet-pretrained MobileNetV2-0.35 (Li et al., 2019), with inputs resized to 96×96 and a classification head consisting of global average pooling, dropout, and a linear layer.

Under the shared training setup, four transfer learning regimes are evaluated, differing only in parameter updates and teacher usage: Linear Probing (LP), where only the classifier is trained; Full Fine-Tuning (FFT), where all parameters are updated; Batch Normalisation Tuning with Linear Probing (BNLP), where only the batch normalisation affine parameters and classifier are trained (Kanavati & Tsuneki, 2021), while running statistics are updated during training; and Logit Standardised Knowledge Distillation (LSKD), where the student is trained via temperature-scaled KL divergence from a fine-tuned ResNet50 teacher (Sun et al., 2024). The teacher is initialised from ImageNet-pretrained weights and fine-tuned on the same PlantVillage data split using the official Sun et al. (2024) configuration, without

additional hyperparameter tuning. Further details of the LSKD formulation are provided in Appendix B.2.

To further validate the training and evaluation pipeline, additional benchmarking was performed against established baselines. Beyond the reproduced MobileNetV2 LP baseline of Seol et al. (2025), this included an LSKD sanity check using the CIFAR-100 ResNet50–MobileNetV2 teacher–student setup of Sun et al. (2024) and an FFT sanity check on Oxford-IIIT Pets. Overall, results were broadly consistent with reported performance (Section 4.1).

Hyperparameters are first explored via a grid search over learning rate ($\text{lr} \in \{0.01, 0.003, 0.001\}$) and dropout probability ($p \in \{0, 0.2, 0.3, 0.5\}$) using three random seeds. Configurations are ranked by mean FP32 validation accuracy, and the top three candidates per regime are re-evaluated across a total of five seeds to assess performance stability. To avoid hyperparameter tuning as a confounding factor, a single unified configuration ($\text{lr} = 0.01, p = 0.5$) is subsequently adopted across all regimes. This configuration provides a controlled comparison of quantisation effects rather than differences arising from training hyperparameters; the unified configuration selection procedure is described in Section 3.4.

3.4 Post-Training Quantisation

Post-training quantisation (PTQ) was performed in PyTorch FX Graph Mode using the standard prepare, calibrate, and convert workflow (Subramanian et al., 2022). Calibration used a fixed class balanced subset of the training data, with 15 images per class (570 total), consistent with prior evidence that several hundred samples are sufficient for reliable PTQ (Nagel et al., 2021). FP32 checkpoints were converted to INT8 using the default FBGEMM backend, with symmetric per-channel weight quantisation and histogram-based per-tensor activation quantisation. Further quantisation details are provided in Appendix B.3.

For a controlled cross-regime comparison, results are reported using a unified training configuration selected from the grid search in Section 3.3, such that each regime remained within 1% of its regime specific INT8 validation grid-based optimum and

showed a broadly similar FP32 to INT8 validation drop. The PTQ pipeline was validated against reference ImageNet-1k INT8 results for ResNet18 and MobileNetV2, with closely matched accuracy and degradation (Section 4.1).

Quantisation performance is reported as mean $\pm$ standard deviation over five seeds for INT8 validation and test accuracy, together with the FP32 to INT8 accuracy drop. Class-level INT8 accuracy is also analysed to assess consistency in class-level difficulty patterns across regimes. Model selection is based solely on INT8 validation accuracy, with the test set reserved for final evaluation and behavioural analysis. Selected behavioural patterns and FP32 to INT8 degradation trends are further checked on a larger disjoint evaluation set.

3.5 Output-Level Prediction Stability Under Quantisation

Accuracy alone does not fully capture whether post-training quantisation (PTQ) preserves predictive behaviour, as high-performing regimes may still differ in how closely their INT8 outputs align with their FP32 counterparts. Consistent with prior behavioural evaluation work (Dutta et al., 2024), we report KL divergence between FP32 and INT8 output distributions to quantify confidence shifts, alongside bit-flip rate, defined as the proportion of samples whose predicted class changes after quantisation, including correct-to-wrong ($C \rightarrow W$) and wrong-to-correct ($W \rightarrow C$) transitions. KL-based alignment losses have also been explored in quantisation research to preserve agreement with full-precision outputs (Liu et al., 2023). Together, these metrics complement accuracy by capturing behavioural consistency, particularly among the stronger regimes with relatively limited degradation.

3.6 Robustness Under Input Corruption

To assess robustness under input corruption, INT8 models are evaluated using controlled corruptions from the ImageNet-C benchmark, specifically Gaussian noise and JPEG compression (Hendrycks & Dietterich, 2019). These corruptions simulate sensor noise and compression artefacts that may be encountered in edge-device deployment scenarios. By progressively increasing corruption severity at inference time, we analyse classification accuracy at each

severity level to assess whether different transfer learning regimes show broadly similar degradation behaviour, despite achieving strong post-quantisation accuracy. All corruptions are applied using the official ImageNet-C implementations to ensure consistency with established robustness evaluation protocols.

4. Results

4.1 Implementation Verification

Prior to the TinyML experiments, we validated backbone initialisation, transfer learning, knowledge distillation, and the FX graph mode PTQ pipeline. The transfer learning setup matched the MobileNetV2 feature extraction (LP) baseline of Seol et al. (2025), with mean validation accuracy across three seeds within 0.12%. Our LSKD implementation was also consistent with the CIFAR-100 LSKD benchmark of Sun et al. (2024), within 0.16% test accuracy in a single-seed comparison using their original setup and dataset. As an additional sanity check, FFT on Oxford-IIIT Pets achieved 86.02% test accuracy, broadly in line with reported MobileNetV2 transfer learning results on this dataset of approximately 88% (Yang et al., 2024; Jang et al., 2025).

The PTQ pipeline was validated independently by applying FX graph mode quantisation to ImageNet-pretrained ResNet18 and MobileNetV2 models and comparing INT8 validation accuracy and degradation against reference values. For ResNet18, results matched the official PyTorch documentation within 0.05%. For MobileNetV2, quantisation produced the expected fourfold model size reduction, with INT8 validation accuracy and degradation within 0.22% and 0.04% of an established PTQ baseline, respectively (Kulkarni et al., 2021). Full details and verification tables are provided in Appendix C.

4.2 Transfer Learning Regime Comparison

Table 1 summarises FP32 and INT8 test performance under the unified configuration (learning rate = 0.01, dropout = 0.5), together with the corresponding FP32 to INT8 accuracy drop. FFT achieves the highest mean INT8 accuracy ($95.42\% \pm 0.84$), followed by LSKD ($94.11\% \pm 0.94$) and BNLP ($93.26\% \pm 1.27$), with degradation ranging from

0.53% to 1.37%. In contrast, LP shows lower INT8 accuracy ($81.58\% \pm 1.29$) and a larger drop (2.68%). Results on the expanded evaluation set (Appendix F.1) show a generally similar overall trend, with LP exhibiting an even larger FP32 to INT8 degradation compared to the other regimes.

Subsequent behavioural analyses therefore focus on FFT, LSKD, and BNLP as the stronger adaptive regimes, while retaining LP as a low-performing reference. Additional experimental details are provided in Appendix D, including scratch baseline, parameter sweeps, and training curves, while Appendix H reports per-class INT8 accuracy and shows broadly overlapping class-level difficulty patterns across regimes.

Table 1: Reports mean test accuracy in FP32 and INT8, together with the corresponding FP32 to INT8 accuracy drop, averaged over five seeds.

Regime	Test Accuracy (%)		
	FP32	INT8	FP32 to INT8 Drop
LP	84.26 ± 0.43	81.58 ± 1.29	2.68 ± 1.31
FFT	95.95 ± 0.40	95.42 ± 0.84	0.53 ± 0.87
BNLP	94.21 ± 1.53	93.26 ± 1.27	0.95 ± 0.92
LSKD	95.47 ± 0.90	94.11 ± 0.94	1.37 ± 1.23

4.3 Output Stability of Model Predictions After Quantisation

Table 2 reports KL divergence and bit-flip statistics on the test set, averaged over five seeds. LP exhibits higher output distribution shift and decision instability under INT8 quantisation, with a mean KL divergence of 0.31 and a bit-flip rate of 13.0%. In contrast, FFT, BNLP, and LSKD exhibit KL divergence in the range of 0.03–0.07 and lower bit-flip rates (1.8–3.6%). FFT attains the lowest mean values among the three regimes although the absolute differences are small. Across all regimes, C→W transitions were more frequent than W→C transitions, consistent with overall accuracy reduction under PTQ. Appendix F.2 shows comparable overall trends on the larger evaluation set.

Table 2: Behavioural deviation between FP32 and INT8 models after post-training quantisation. KL divergence measures output distribution shift, while bit-flip rate reports prediction changes, including correct-to-wrong (C→W) and wrong-to-correct (W→C) transitions; wrong-to-wrong transitions are omitted for brevity. Values are mean $\pm$ standard deviation over five seeds.

Regime	KL Divergence	Bit Flip (%)	C→W (%)	W→C (%)
LP	0.31 ± 0.03	13.00 ± 1.39	6.42 ± 1.44	3.74 ± 0.60
FFT	0.03 ± 0.01	1.84 ± 0.77	1.05 ± 0.89	0.53 ± 0.19
BNLP	0.05 ± 0.01	3.58 ± 0.40	2.05 ± 0.68	1.11 ± 0.39
LSKD	0.07 ± 0.02	3.26 ± 0.86	2.16 ± 0.92	0.79 ± 0.62

4.4 Behaviour Under Corruption

All transfer learning regimes exhibit progressive accuracy degradation as corruption severity increases. Under Gaussian noise (Figure 1), FFT and LSKD remain closely aligned, with LSKD slightly higher at severe noise levels. BNLP is modestly lower at low-to-moderate noise, consistent with its slightly lower clean INT8 accuracy among the stronger regimes, but converges towards LSKD at higher severities, where error bars overlap. LP is lower at low noise but achieves comparatively higher mean accuracy under stronger noise. Under JPEG compression (Figure 2), trends are largely aligned at low-to-moderate severities, with clearer separation emerging only at higher compression strengths, where LP and BNLP fall below FFT and LSKD. Appendix G shows similar trends for macro-F1, tabulated results, and the larger evaluation set; macro-F1 follows the same overall pattern but is typically 2–5% lower than accuracy at higher severities for Gaussian noise.

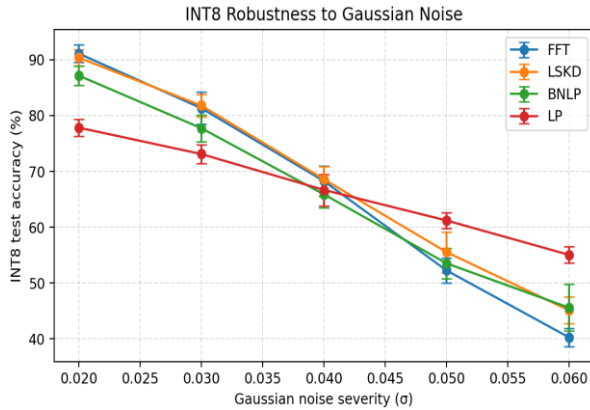


Figure 1: INT8 test accuracy under Gaussian noise corruption ($\sigma = 0.02$ – 0.06) for each transfer learning regime. Points denote mean accuracy over five seeds; error bars indicate standard deviation.

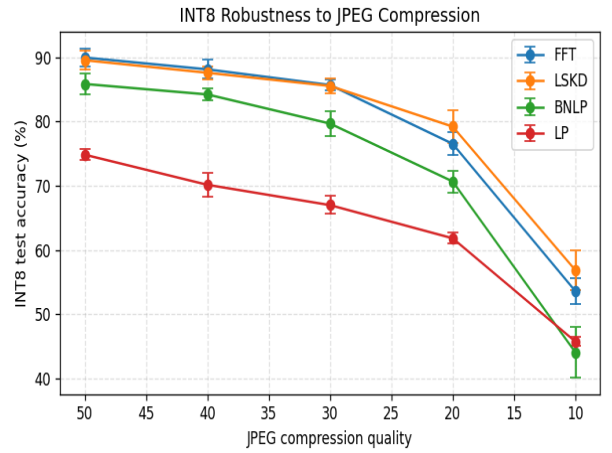


Figure 2: INT8 test accuracy under JPEG Compression (quality = 50–10) for each transfer learning regime. Points denote mean accuracy over five seeds; error bars indicate standard deviation.

5. Discussion

5.1 Transfer Learning Regimes Performance

Table 1 shows that multiple transfer learning regimes achieve strong post-quantisation accuracy despite differences in training cost and parameter updates. Methods that update all network parameters (FFT and LSKD) attain the highest FP32 and INT8 accuracy, while BNLP offers a favourable accuracy-efficiency trade-off, achieving modestly lower post-quantisation performance while updating only batch-normalisation parameters and the classifier head. This suggests that BNLP may serve as a practical alternative when training resources or full retraining pipelines are constrained. In contrast, LP exhibits lower INT8 accuracy and a larger quantisation-induced drop.

This weaker performance is consistent with established findings on feature transferability: early layers tend to learn general features, whereas later layers become increasingly task-specific (Yosinski et al., 2014). By freezing the backbone, linear probing restricts adaptation in deeper layers, which may limit task-specific specialisation under limited data. In line with broader benchmarks, fine-tuning typically achieves higher in-distribution accuracy than approaches with a frozen backbone (Kornblith et al., 2019; Kumar et al., 2022), consistent with our observation that LP attains the lowest FP32 accuracy while regimes permitting backbone adaptation perform more strongly.

Within these adaptive approaches, partial adaptation through BNLP remains competitive with FFT, reflecting prior evidence that updating only batch normalisation affine parameters may capture much of the benefit of full backbone adaptation (Kanavati & Tsuneki, 2021). A plausible explanation is that updating these affine parameters allows intermediate features to be rescaled and shifted toward the target task without modifying the convolutional weights. However, incorporating LSKD yields no clear improvement over FFT in this setting, despite the stronger teacher achieving 97.89% test accuracy, suggesting that distillation gains may depend on teacher-student capacity alignment and can diminish under mismatch (Cho & Hariharan, 2019). Here, the teacher network (ResNet50) is substantially larger than the MobileNetV2-0.35 student, which may limit the student’s ability to fully approximate the teacher’s decision boundaries.

The observed FP32 to INT8 degradation falls within ranges reported for MobileNetV2 in prior ImageNet based-PTQ studies, approximately 0.5–5% depending on configuration (Nagel et al., 2021; Kulkarni et al., 2021). However, variation across transfer learning regimes suggests that quantisation sensitivity is not purely architecture dependent. LP shows the largest degradation, whereas FFT, BNLP, and LSKD remain broadly comparable overall, with FFT yielding the lowest mean degradation. Notably, BNLP updates only batch normalisation parameters yet has degradation patterns closer to the adaptive regimes than to LP, suggesting that even limited backbone adaptation can matter for post-training quantisation sensitivity in this context. One possible explanation

is that regimes which adapt the backbone during transfer learning produce activations during calibration that are more representative of the target dataset, which may in turn influence the estimated quantisation parameters. Results on the expanded evaluation set (Appendix F.1) show the same overall trend, with LP exhibiting an even larger degradation, while LSKD shows a slightly higher drop than FFT. Overall, these findings suggest that PTQ sensitivity may depend not only on whether the backbone is adapted, but also on how it is adapted during transfer learning.

5.2 Post-Quantisation Behavioural Stability and Corruption Robustness

Output-level behavioural metrics assessed whether broadly similar INT8 degradation among the stronger regimes corresponded to a comparable alignment with their FP32 counterparts. In this setting, regimes with higher INT8 accuracy and relatively small FP32 to INT8 degradation generally show lower KL divergence and bit-flip rates, indicating better preservation of output distributions under quantisation. Among the stronger regimes, FFT has the lowest mean KL divergence and bit-flip rate, although differences relative to BNLP and LSKD remain small. By contrast, LP shows higher KL divergence and bit-flip rates, consistent with its larger FP32 to INT8 accuracy drop. Overall, these results suggest that the magnitude of FP32 to INT8 accuracy degradation may capture much of the observed output-level behavioural instability, with only minor secondary differences across the stronger regimes.

Corruption-based evaluation shows broadly similar degradation patterns across the stronger regimes. FFT, LSKD, and BNLP show closely aligned trends at low-to-moderate corruption levels, with some differences emerging only at higher severities. Under severe Gaussian noise, LP retains higher accuracy despite lower clean performance. This is consistent with Kumar et al. (2022), who suggest that LP may better preserve generic pretrained representations, whereas broader backbone adaptation may distort them through task-specific adaptation. In our setting, this may plausibly help explain LP’s relative advantage under severe Gaussian noise, where generic pretrained representations may be more useful, although the same advantage does not appear uniformly across corruption types. Overall, robustness

differences across regimes become more visible mainly under severe corruption levels.

Supplementary class-level behavioural analyses (Appendix H) show overlap in the set of low-performing classes across regimes. Within this dataset, this pattern suggests that post-quantisation class-level performance may relate more to class-specific data difficulty than to the choice of transfer learning strategy, although the magnitude of class-wise accuracy can still differ across regimes.

5.3 Overall Implications

From a practical perspective, FFT and LSKD achieve the highest overall INT8 performance in this study, while BNLP offers a more parameter-efficient alternative when training resources are constrained, with only modest reductions in accuracy. Although Knowledge Distillation can be appealing because it uses a larger teacher model, the results suggest that it does not yield a clear benefit in every setting and may also increase training cost when an additional teacher must be trained or fine-tuned. Taken together, these results suggest that, in this setting, aggregate metrics captured most of the practical differences among the stronger regimes.

From a broader research perspective, these results suggest that PTQ studies conducted under a single adaptive transfer learning regime may still capture the main degradation and behavioural trends among the stronger regimes under the conditions examined. However, minor regime-specific differences may still be missed, while larger differences may emerge when the backbone is kept fully frozen.

5.4 Limitations and Future Work

This study is limited to a single architecture (MobileNetV2-0.35) and dataset (PlantVillage), and behavioural patterns may differ for larger models, alternative TinyML architectures, or datasets with greater visual diversity. Direct numerical comparison with prior PTQ literature is also constrained, as most reported benchmarks use datasets different from the task-specific agricultural dataset used here. In addition, robustness evaluation is restricted to Gaussian noise and JPEG compression, and the analysis focuses on post-training quantisation using a fixed calibration and observer configuration. Future

work could extend this by evaluating additional corruption types, alternative calibration strategies, different observer configurations, and quantisation-aware training.

Future work could also examine the structural origins of quantisation sensitivity. Rather than comparing transfer learning regimes only at the aggregate level, layer-wise or block-wise analyses could identify where quantisation error arises within the network, and which components are most sensitive. This would help clarify whether such sensitivity is driven primarily by the transfer learning regime, the architecture, or their interaction.

6. Conclusion

This study examined how post-training quantisation interacts with different transfer learning regimes in a TinyML setting using MobileNetV2-0.35. Across the evaluated regimes, the most apparent post-quantisation difference was observed for Linear Probing, which showed lower INT8 performance, greater FP32 to INT8 degradation, and larger output-level behavioural deviation than the stronger adaptive regimes. Within this stronger group, Full Fine-Tuning, Batch Normalisation Tuning, and Logit Standardised Knowledge Distillation remained broadly similar overall, with behavioural diagnostics showing largely aligned patterns and only some differences emerging under severe corruption. Full Fine-Tuning showed the lowest output-level deviation and degradation, although differences remained small overall.

In summary, the results indicate that PTQ degradation and post-quantisation behaviour were broadly similar across the stronger adaptive regimes, while larger differences emerged when the backbone was kept frozen. Aggregate performance metrics appeared to capture most of the practical post-quantisation differences observed in this study, while behavioural diagnostics provided additional detail without materially changing the general pattern across the stronger regimes.

Data and Code Availability

To support reproducibility, the [project repository](#) provides the source code and includes links to model checkpoints, supporting datasets and additional files.

7. References

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8. Appendix

Appendix A: Literature Review

A.1 Knowledge Distillation

Knowledge Distillation (KD) trains a student model under the guidance of a pretrained teacher model by encouraging the student to match the teacher's softened output distribution in addition to the ground-truth labels (Hinton et al., 2015).

Let z_s and z_t denote the student and teacher logits respectively. Temperature-scaled probabilities are defined as

$$p_t^{(T)} = \text{softmax}(z_t/T) \text{ and } p_s^{(T)} = \text{softmax}(z_s/T)$$

Where $T > 1$ is a temperature parameter that produces a softer probability distribution by reducing confidence peaks.

The total training objective combines standard cross-entropy with a distillation term:

$$L = aL_{CE}(y, \hat{y}_s) + bKL(p_t^{(T)}, p_s^{(T)}),$$

where

- L_{CE} is the cross-entropy loss between ground-truth labels y and student predictions $\hat{y}_s$
- $KL(p_t^{(T)}, p_s^{(T)})$ is the Kullback-Leibler divergence between the temperature-scaled teacher and student probability distributions. It measures how closely the student's softened output distribution matches that of the teacher; smaller values indicate better alignment.
- a and b weight the relative contributions of the hard-label and distillation terms
- T controls the softness of the distillation distributions

The distillation term encourages the student model to match the teacher's softened class probabilities, transferring inter-class relational information beyond the ground-truth signal.

A.2 Batch Normalisation and Folding

MobileNetV2 is a lightweight convolutional architecture designed for resource constrained environments. It consists of stacked inverted residual blocks built from depthwise separable convolutions, where each convolutional layer is followed by batch normalisation. As a result, batch normalisation (BN) is applied repeatedly throughout the network and plays a central role in feature scaling and representation learning. Since this study evaluates batch normalisation tuning and post-training quantisation, understanding the behaviour of BN layers in MobileNetV2 is important.

A.2.1 Batch Normalisation

Batch Normalisation (BN) normalises intermediate activation to stabilise training (Ioffe & Szegedy, 2015). For a given channel, activations are normalised using mini-batch mean and variance statistics

$$y = \gamma \frac{x - \mu_B}{\sqrt{\sigma_B^2 + \epsilon}} + \beta$$

where μ_B and σ_B are the batch mean and variance respectively, and γ (scale), β (shift) are learnable affine parameters. The affine parameters allow the network to rescale and shift activations after normalisation, so the model can still learn useful feature representations.

During training, the mean and variance of the activations are computed for each mini batch. Running estimates of these statistics are accumulated across batches and used during inference to ensure deterministic behaviour. In this study, BNLP updates only the affine parameters and the running statistics, while convolutional weights remain frozen. This allows limited adaptation of feature scaling without modifying the underlying convolutional filters.

A.2.2 Batch Normalisation Folding

For deployment and post-training quantisation, BN layers are typically folded into the preceding convolutional layer, by absorbing the affine scaling and shifting terms into the convolution weights and bias. This removes BN as a separate operation during inference for better latency. Folding modifies the effective weight magnitudes:

$$W' = W \cdot \frac{\gamma}{\sqrt{\sigma^2 + \epsilon}}$$

which can alter weight distributions and influence quantisation scale estimation. Since PTQ depends on observed weight ranges, BN folding directly affects quantisation behaviour (Krishnamoorthi, 2018).

Appendix B: Methodology

This appendix provides additional detail on the methodology used in this study.

B.1 Training Setup Summary

Table B.1. Summarises the setup used in this study for the unified configuration highlighted in section 3.3.

Component	Setting
Backbone	MobileNetV2-0.35 (ImageNet pretrained; Li et al., 2019)
Input resolution	96×96
Classifier head	Global Average Pooling $\rightarrow$ Dropout $\rightarrow$ Linear (38 classes)
Optimiser	Stochastic Gradient Descent with Momentum (SGDM)
Momentum	0.9
Batch size	50

Learning rate (unified)	0.01
Dropout probability (unified)	0.5
Maximum epochs	200
Early stopping	Patience = 10 (validation accuracy)
LR scheduler	ReduceLROnPlateau (patience = 5, factor = 0.1)
Loss function	Cross-entropy (LP, FFT, BN-LP); KD loss for LSKD
Number of seeds	3 (hyperparameter search), 5 (final evaluation)
Seeds Used	0,2,5 (hyperparameter search) and 0,2,5,42,1337 (final evaluation)

B.2 LSKD Implementation Detail

In this work, we follow Sun et al. (2024), where both student and teacher logits are standardised prior to temperature scaling. Specifically, for a logit vector z , standardised logits are computed as:

$$\hat{z} = \frac{z - \mu(z)}{\sigma(z)}$$

where $\mu(z)$ and $\sigma(z)$ denote mean and standard deviation of the logits. The standardised logits are then passed through temperature-scaled SoftMax before computing the KL divergence. This procedure reduces sensitivity to the logit magnitude differences and encourages students to focus on relative class relationships rather than absolute score scale. Appendix A.1 provides general KD formulation.

B.3 Post-Training Quantisation

This project uses post-training quantisation implemented through PyTorch FX graph mode with the FBGEMM backend. The formulation below summarises the uniform affine quantisation scheme and relates it to the implementation used in this study.

B.3.1 Quantisation Formula

Given a real-valued tensor $x \in R^n$ affine quantisation maps values to an integer grid defined by a bit width, scale factor $s > 0$ derived from the value range (determines quantisation step size), and a zero-point z . Quantisation and de-quantisation are defined as:

$$\begin{aligned} x_{int} &= \text{clamp}(\text{round}(x/s) + z, q_{min}, q_{max}), \\ \hat{x} &= s(x_{int} - z) \end{aligned}$$

where q_{min} and q_{max} define the valid integer range. The zero-point z ensures that the real value zero is represented exactly on the integer grid. This is important because many neural network operations, such as ReLU activations, rely on exact zeros. In the symmetric case used for weights, the zero point is fixed to zero, simplifying the mapping between floating point and integer values (Nagel et al., 2021).

For activations, the parameters (s , z) are estimated from calibration data using a histogram-based observer that collects a distribution of activation values during calibration and selects an appropriate quantisation range based on that distribution. Activations are quantised per-tensor using the affine formulation above, where values are scaled, rounded, shifted by the zero-point, and clamped to the valid 8-bit range. For weights, symmetric ($z = 0$) per-channel 8-bit quantisation is used, where a separate scale factor is computed for each output channel to reduce relative quantisation error across channels. The scale is determined using the per-channel min-max values.

Appendix C: Experimental Pipeline Verification

This appendix supports the correctness of the training, distillation, and quantisation pipelines used in this study. These checks ensure consistency with prior work before conducting the main PlantVillage experiments.

C.1 Training Pipeline Verification

We reproduce the MobileNetV2 feature-extraction (LP) baseline of Seol et al. (2025) on the Li and Zhao (2019) crack dataset. Using the same backbone initialisation and feature-extraction protocol, we obtain a mean validation accuracy (across 3 random seeds) of 97.34%, closely matching the reported 97.46%. Table C.1 shows accuracy, precision, and recall, demonstrating close agreement across all metrics.

Table C.1. Comparison of performance metrics between reproduced results and Seol et al.

Metric	Ours	Seol’s	Difference (Ours - Seol’s)
Accuracy	97.34	97.46	-0.12
Precision	97.33	96.43	+0.90
Recall	97.32	98.57	-1.25

We further evaluate the unified training configuration on the Oxford-IIIT Pets dataset, which provides an official test split. Using $LR = 1e-2$, dropout = 0.5, and seed 0, MobileNetV2-1.0 achieves 86.02% test accuracy under Full Fine-Tuning. This lies within the range typically reported for MobileNetV2 transfer learning on this dataset ($\approx 85-88\%$), supporting correct end-to-end training behaviour.

Table C.2. Comparison of FP32 test accuracy on Oxford-IIIT Pets. Reported values from prior work correspond to the best-performing configuration in each study; differences may reflect variation in transfer learning strategy and overall training setup.

Model	FP32 Test Accuracy (%)
FFT (Ours)	86.02
Yang et al. (2024)	87.98
Jang et al. (2025)	88.06

Additional single-run checks were used to validate backbone initialisation and the implementation of Logit Standardisation Knowledge Distillation (LSKD). The ImageNet-pretrained MobileNetV2-0.35 backbone achieves 60.10% Top-1 and 82.19% Top-5 on ImageNet-1k validation, consistent with published values. Reproducing the CIFAR-100 ResNet50-MobileNetV2 teacher-student setup of Sun et al. (2024), across one seed, yields 69.18% test accuracy (reported: 69.02%), indicating close agreement and supporting the correctness of the LSKD implementation.

C.2 Quantisation Pipeline Verification

We validate our FX graph-mode PTQ implementation using an ImageNet-pretrained ResNet18. Pre-processing follows the standard ImageNet validation pipeline. Calibration uses 1,024 training images; evaluation uses the full validation split. FP32 and INT8 validation accuracies closely match PyTorch reference results. Table C.3 summarises these comparisons.

Table C.3. Verification of ResNet18 FP32 and INT8 validation accuracy against PyTorch reference implementation.

Metric	Ours	PyTorch’s Reference	Difference (Ours - PyTorch’s)
FP32 Validation Accuracy (%)	69.75	69.76	-0.01
INT8 Validation Accuracy (%)	69.44	69.49	-0.05
FP32 to INT8 Validation Accuracy Drop	0.31	0.27	+0.04
File Size (MB)	11.36	11.20	+0.16

We also compare MobileNetV2 PTQ behaviour against a previously reported result (Kulkarni et al., 2021). In that study, standard PTQ produced an accuracy drop of 4.19%, while our FX graph-mode implementation produced a 4.15% drop, indicating broadly similar behaviour.

Table C.4. Comparison of FP32 and INT8 validation accuracy against prior reported results.

Metric	Ours	Kulkarni et al. (2021)	Difference (Ours - Kulkarni)
FP32 Validation Accuracy (%)	72.18	72.00	+0.18
INT8 Validation Accuracy (%)	68.03	67.81	+0.22
FP32 to INT8 Validation Accuracy Drop	4.15	4.19	-0.04
Method	PyTorch	TensorFlow	-

For MobileNetV2-0.35, state_dict size reduces from 1.85 MB to 0.64 MB ($\sim 2.9\times$ reduction). When considering only raw tensor storage, the size decreases from 1.75 MB to 0.39 MB ($\sim 4\times$ reduction), consistent with the expected compression from 32-bit to 8-bit parameters. Inspection of the converted FX graph confirms correct replacement of convolutional and linear layers with quantised modules.

These results indicate that the unified training pipeline produces results broadly comparable to established baselines under both feature-extraction and Full Fine-Tuning regimes, as well as under quantisation. The same pipeline is used consistently across all PlantVillage experiments.

Appendix D: Transfer Learning and Quantisation Results from Grid-Selected Candidate Configurations

This appendix reports results from the hyperparameter grid search conducted to identify the strongest configuration per transfer learning regime and to justify the unified configuration used in the main analysis. Training curves (for the unified configuration) and a scratch-training baseline are also included for reference.

D.1 Method Sweep

This section summarises validation accuracy results from the hyperparameter sweep for each transfer learning regime. Results are reported as mean $\pm$ standard deviation across three random seeds.

D.1a Full Fine-Tuning (FFT)

Table D.1. Hyperparameter sweep for FFT. Mean $\pm$ standard deviation over three random seeds.

Config ID	LR	Dropout	Validation Accuracy (%)
1	1e-2	0	95.31% $\pm$ 0.65%
2	1e-2	0.2	94.56% $\pm$ 0.76%
3	1e-2	0.3	94.52% $\pm$ 0.73%
4	1e-2	0.5	95.70% $\pm$ 0.95%
5	1e-3	0	94.25% $\pm$ 0.54%
6	1e-3	0.2	94.17% $\pm$ 0.19%
7	1e-3	0.3	94.74% $\pm$ 0.60%
8	1e-3	0.5	94.52% $\pm$ 0.20%
9	3e-3	0	94.39% $\pm$ 0.55%
10	3e-3	0.2	94.43% $\pm$ 0.53%
11	3e-3	0.3	95.31% $\pm$ 0.38%
12	3e-3	0.5	95.22% $\pm$ 0.62%

D.1b Linear Probing (LP)

Table D.2. Hyperparameter sweep for LP. Mean $\pm$ standard deviation over three random seeds.

Config ID	LR	Dropout	Validation Accuracy (%)
1	1e-2	0	85.79 $\pm$ 0.22
2	1e-2	0.2	86.05 $\pm$ 0.13
3	1e-2	0.3	85.44 $\pm$ 0.30
4	1e-2	0.5	84.72 $\pm$ 0.80
5	1e-3	0	84.43 $\pm$ 0.20
6	1e-3	0.2	83.55 $\pm$ 1.17
7	1e-3	0.3	83.82 $\pm$ 1.42
8	1e-3	0.5	83.11 $\pm$ 1.22
9	3e-3	0	85.44 $\pm$ 0.65
10	3e-3	0.2	85.13 $\pm$ 0.23
11	3e-3	0.3	84.87 $\pm$ 1.94
12	3e-3	0.5	84.37 $\pm$ 1.12

D.1c Batch Normalisation Tuning with Linear Probing (BNLP)

Table D.3. Hyperparameter sweep for BNLP. Mean $\pm$ standard deviation over three random seeds.

Config ID	LR	Dropout	Validation Accuracy (%)
1	1e-2	0	93.25% $\pm$ 0.66%
2	1e-2	0.2	93.86% $\pm$ 0.28%
3	1e-2	0.3	93.42% $\pm$ 0.95%
4	1e-2	0.5	94.41% $\pm$ 0.79%
5	1e-3	0	93.12% $\pm$ 0.15%
6	1e-3	0.2	92.94% $\pm$ 0.33%
7	1e-3	0.3	92.85% $\pm$ 0.99%
8	1e-3	0.5	92.72% $\pm$ 0.20%
9	3e-3	0	93.90% $\pm$ 0.38%

10	3e-3	0.2	93.99% $\pm$ 0.28%
11	3e-3	0.3	94.08% $\pm$ 0.26%
12	3e-3	0.5	93.55% $\pm$ 1.36%

D.1d Logit-Standardisation Knowledge Distillation (LSKD)

The LSKD regime introduces two additional hyperparameters: distillation temperature and distillation weighting. Following Sun et al. (2024), the cross-entropy weight was fixed at 0.1, and three temperature-weight pairs were evaluated: (3,1), (9,2), and (12,4). Raw sweep results are omitted for brevity; shortlisted candidates are presented in Section D.2.

The teacher model was trained using Sun et al.’s original procedure, and one fixed teacher checkpoint was used for all distillation experiments, achieving 98.55% validation accuracy.

D.2 Grid-Selected Candidate Configuration Results

For each regime, the top three configurations from the initial sweep were shortlisted. Each candidate was evaluated over two additional seeds (five total), and corresponding INT8 models were produced via post-training quantisation. The configuration with the highest mean INT8 validation accuracy was selected.

Tables D.4–D.7 report FP32 validation accuracy, INT8 validation accuracy, and FP32→INT8 accuracy drop for each candidate.

D.1a Full Fine-Tuning (FFT)

Table D.4. Shortlisted FFT candidate configurations evaluated over five seeds. Mean $\pm$ standard deviation reported for FP32 and INT8 validation accuracy.

Metric	Candidate A (LR = 0.01, Dropout = 0.0)	Candidate B (LR = 0.01, Dropout = 0.5)	Candidate C (LR = 0.003, Dropout = 0.3)
FP32 Validation Accuracy (%)	94.97 $\pm$ 0.65	95.61 $\pm$ 0.69	95.08 $\pm$ 0.41
INT8 Validation Accuracy (%)	93.81 $\pm$ 1.10	94.61 $\pm$ 0.64	92.63 $\pm$ 1.40
Validation Accuracy Drop (%)	1.16 $\pm$ 0.67	1.00 $\pm$ 0.30	2.45 $\pm$ 1.29

D.1b Linear Probing (LP)

Table D.5. Shortlisted LP candidate configurations evaluated over five seeds.

Metric	Candidate A (LR = 0.01, Dropout = 0)	Candidate A (LR = 0.01, Dropout = 0.2)	Candidate B (LR = 0.01, Dropout = 0.3)
FP32 Validation Accuracy (%)	85.76 $\pm$ 0.17	86.03 $\pm$ 0.17	85.47 $\pm$ 0.60
INT8 Validation Accuracy (%)	80.26 $\pm$ 0.54	80.66 $\pm$ 0.90	79.13 $\pm$ 1.66
Validation Accuracy Drop (%)	5.50 $\pm$ 0.43	5.37 $\pm$ 0.98	6.34 $\pm$ 1.20

D.1c Batch Normalisation Tuning with Linear Probing (BNLP)

Table D.6. Shortlisted BNLP candidate configurations evaluated over five seeds.

Metric	Candidate A (LR = 0.01, Dropout = 0.5)	Candidate B (LR = 0.003, Dropout = 0.2)	Candidate C (LR = 0.003, Dropout = 0.3)
FP32 Validation Accuracy (%)	94.08 $\pm$ 0.56	93.87 $\pm$ 0.38	93.89 $\pm$ 0.45
INT8 Validation Accuracy (%)	93.18 $\pm$ 0.71	91.63 $\pm$ 0.66	92.23 $\pm$ 0.49
Validation Accuracy Drop (%)	0.90 $\pm$ 0.63	2.24 $\pm$ 0.58	1.66 $\pm$ 0.66

D.1d Logit-Standardisation Knowledge Distillation (LSKD)

Table D.7. Shortlisted LSKD candidate configurations evaluated over five seeds.

Metric	Candidate A (LR = 0.01, Dropout = 0.0, Temperature = 3, KD Weighting = 1)	Candidate B (LR = 0.01, Dropout = 0.0, Temperature = 12, KD Weighting = 4)	Candidate A (LR = 0.01, Dropout = 0.2, Temperature = 12, KD Weighting = 4)
FP32 Validation Accuracy (%)	96.45 $\pm$ 0.38	96.37 $\pm$ 0.44	96.47 $\pm$ 0.47
INT8 Validation Accuracy (%)	94.81 $\pm$ 0.71	95.00 $\pm$ 0.87	95.13 $\pm$ 0.45
Validation Accuracy Drop (%)	1.64 $\pm$ 0.55	1.37 $\pm$ 0.52	1.34 $\pm$ 0.11

D.3 Results of Grid-Selected Configurations

Table D.8 summarises validation and test performance for the configuration selected per regime based on highest mean INT8 validation accuracy. Table D.9 reports the INT8 accuracy difference between the unified configuration and the best-in-grid configuration (Unified – Best). This comparison quantifies the trade-off introduced by enforcing a shared configuration across regimes.

Table D.8. Performance of grid-selected configuration per regime. Mean $\pm$ standard deviation over five seeds. Drop denotes FP32 $\rightarrow$ INT8 accuracy reduction.

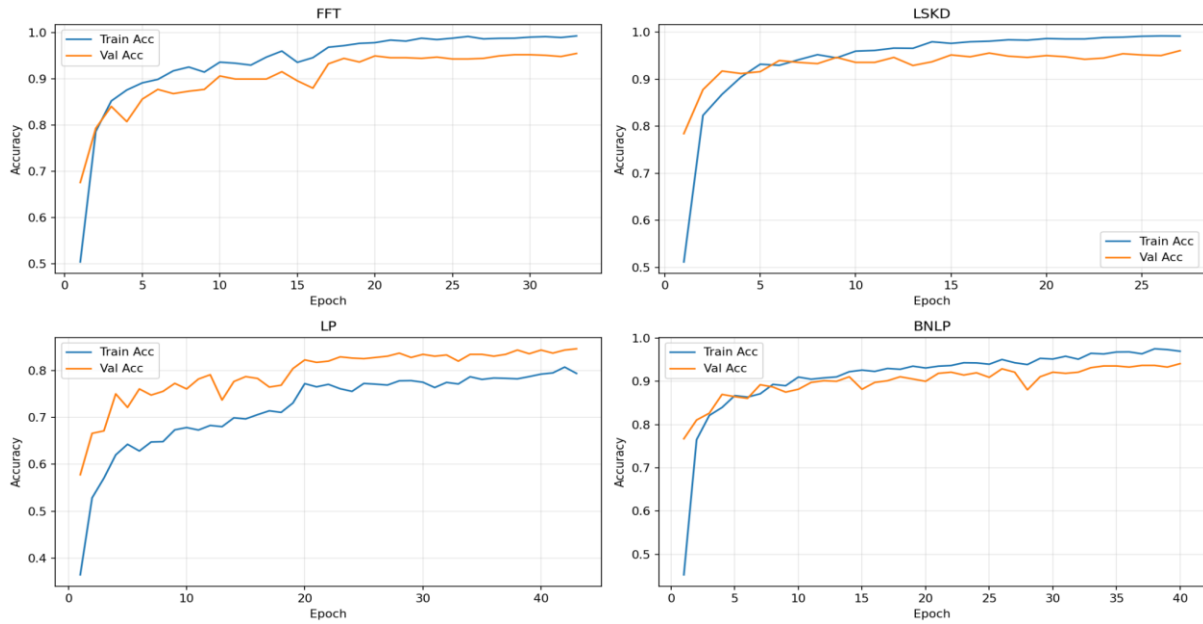
Regime	Validation Accuracy (%)			Test Accuracy (%)		
	FP32	INT8	Drop (FP32 $\rightarrow$ INT8)	FP32	INT8	Drop (FP32 $\rightarrow$ INT8)
LP (LR = 0.01, Dropout = 0.2)	86.03 $\pm$ 0.17	80.66 $\pm$ 0.90	5.37 $\pm$ 0.98	86.00 $\pm$ 0.92	81.37 $\pm$ 1.18	4.63 $\pm$ 1.79
FFT (LR = 0.01, Dropout = 0.5)	95.61 $\pm$ 0.69	94.61 $\pm$ 0.64	1.00 $\pm$ 0.30	95.95 $\pm$ 0.40	95.42 $\pm$ 0.84	0.53 $\pm$ 0.87
BNLP (LR = 0.01, Dropout = 0.5)	94.08 $\pm$ 0.56	93.18 $\pm$ 0.71	0.90 $\pm$ 0.63	94.21 $\pm$ 1.53	93.26 $\pm$ 1.27	0.95 $\pm$ 0.92
LSKD (LR = 0.01, Dropout = 0.2)	96.47 $\pm$ 0.47	95.13 $\pm$ 0.45	1.34 $\pm$ 0.11	95.63 $\pm$ 0.40	94.42 $\pm$ 1.09	1.21 $\pm$ 1.05

Table D.9. INT8 accuracy and FP32 to INT8 drop difference between unified configuration and best-in-grid configuration (Unified – Best).

Regime	INT8 Validation Accuracy Difference	Validation FP32 to INT8 Drop Difference	INT8 Test Accuracy Difference
LP	-0.90	-0.45	+0.21
FFT	0.00	0.00	0.00
BNLP	0.00	0.00	0.00
LSKD	-0.45	+0.05	-0.31

D.4 Training Curves

Figure D.1. Training curves obtained by each method under the unified configuration.



D.5 Scratch Result

A scratch-training baseline was evaluated using the unified training setup with additional regularisation (weight decay = 5×10^{-4}). Only limited sanity checks were performed rather than extensive hyperparameter optimisation. Scratch models exhibited a persistent generalisation gap, consistent with the dataset being label-limited when training from random initialisation. Under the same setup, transfer learning regimes achieved substantially stronger generalisation. This baseline is included for contextual comparison only and is not part of the primary analysis.

Table D.10. Comparison of scratch training and transfer learning regimes under the unified training configuration. Mean $\pm$ standard deviation over five seeds.

Training Regime	FP32 Val Accuracy (%)	FP32 Test Accuracy (%)
Scratch	78.16 ± 0.50	78.21 ± 1.76
LP	84.68 ± 0.77	84.26 ± 0.43
FFT	95.61 ± 0.69	95.95 ± 0.40
BNL	94.08 ± 0.56	94.21 ± 1.53
LSKD	96.08 ± 0.39	95.47 ± 0.90

Appendix E: Calibration Set

This appendix evaluates the effect of calibration set size on INT8 validation accuracy across transfer learning regimes. The goal is to determine whether calibration size materially influences quantised performance and to justify the fixed calibration size used in subsequent experiments.

E.1 Effect of Calibration Set Size

Table E.1 reports INT8 validation accuracy (mean $\pm$ standard deviation across five seeds) for calibration sizes of 5, 10, 15, and 20 images per class. Across regimes, increasing calibration size does not yield consistent improvements in INT8 accuracy. FFT, BNLP, and LSKD remain generally stable across calibration sizes. LP exhibits greater variability, indicating higher sensitivity to calibration subset composition. Overall, these results suggest that once a small representative calibration set is used, further increases in calibration size provide limited benefit.

Table E.1. INT8 validation accuracy under varying calibration sizes (mean $\pm$ standard deviation over five seeds).

Regime	INT8 Validation Accuracy (N=5)	INT8 Validation Accuracy (N=10)	INT8 Validation Accuracy (N=15)	INT8 Validation Accuracy (N=20)
LP	80.76 $\pm$ 1.06	77.79 $\pm$ 0.86	79.76 $\pm$ 0.63	78.63 $\pm$ 0.68
FFT	94.61 $\pm$ 0.77	94.82 $\pm$ 0.80	94.61 $\pm$ 0.64	94.97 $\pm$ 0.64
BNLP	92.82 $\pm$ 1.16	93.11 $\pm$ 0.85	93.18 $\pm$ 0.71	92.84 $\pm$ 0.96
LSKD	94.16 $\pm$ 0.51	94.82 $\pm$ 0.36	94.68 $\pm$ 0.42	94.45 $\pm$ 0.41

E.2 Choosing Calibration Set

Based on the stability observed in Table E.1, a fixed calibration size of N = 15 images per class is adopted for all subsequent experiments. Reducing calibration to N = 5 introduces modest degradation for BNLP and LSKD. Increasing calibration to N = 20 provides only marginal or inconsistent changes across regimes. Overall, N = 15 offers a balanced trade-off between calibration stability and efficiency. Table E.2 reports the accuracy difference between N = 15 and alternative calibration sizes.

Table E.2. INT8 validation accuracy difference relative to N = 15 (mean over five seeds).

Regime	INT8 Validation Accuracy Difference (N = 5 - N = 15)	INT8 Validation Accuracy Difference (N = 10 - N = 15)	INT8 Validation Accuracy Difference (N = 20 - N = 15)
LP	+1.00	-1.97	-1.13
FFT	+0.00	+0.21	+0.36
BNLP	-0.36	-0.07	-0.34
LSKD	-0.52	+0.14	-0.23

Appendix F: Increased Dataset and KL Divergence

This appendix evaluates model performance and output-level behavioural metrics on an expanded evaluation set (≈ 100 images per class) to assess robustness of the reported findings.

F.1 Accuracy Results on the Expanded Evaluation Set

Table F.1 reports INT8 test accuracy and INT8 F1 score (mean $\pm$ standard deviation across five seeds) on the expanded evaluation set. Absolute INT8 performance decreases slightly across all regimes relative to the smaller test split, with the larger and more diverse evaluation set providing a more conservative estimate of generalisation. Importantly, LP shows an even larger accuracy drop compared to other regimes. Moreover, LSKD shows a minor increase in degradation compared to the other stronger regimes, although the absolute differences remain small.

Table F.1. INT8 test accuracy and FP32 to INT8 accuracy drop on the expanded evaluation set (mean $\pm$ standard deviation over five seeds).

Regime	INT8 Test Accuracy (%)	FP32 to INT8 Test Accuracy Drop (%)
LP	78.88 ± 0.65	4.71 ± 0.40
FFT	94.44 ± 0.40	0.55 ± 0.13
BNLP	91.93 ± 0.55	0.62 ± 0.32
LSKD	93.70 ± 0.55	1.28 ± 0.28

F.2 Behavioural Stability Results on the Expanded Evaluation Set

Table F.2 reports behavioural metrics computed on the expanded evaluation set, including KL divergence, overall bit-flip rate, correct-to-wrong transitions ($C \rightarrow W$), and wrong-to-correct transitions ($W \rightarrow C$). The behavioural patterns are consistent with those reported in Section 4.2.

Table F.2. Behavioural metrics on the expanded evaluation set (mean $\pm$ standard deviation over five seeds).

Regime	KL Divergence	Bit Flip (%)	$C \rightarrow W$ (%)	$W \rightarrow C$ (%)
LP	0.32 ± 0.02	15.72 ± 0.67	8.57 ± 0.22	3.86 ± 0.30
FFT	0.03 ± 0.00	2.23 ± 0.35	1.24 ± 0.17	0.70 ± 0.10
BNLP	0.05 ± 0.01	3.73 ± 0.61	1.90 ± 0.44	1.29 ± 0.21
LSKD	0.07 ± 0.01	3.30 ± 0.54	2.09 ± 0.39	0.81 ± 0.16

Appendix G: Extended Input Perturbation Analysis

This appendix provides additional robustness results supporting Section 4.4, including evaluations on the expanded set. Across corruption types, regimes behave generally similar at low-to-moderate severities, with some separation emerging at higher severities. The primary trends reported in Section 4.4 remain consistent.

G.1 Gaussian Noise Table Results

Tables G.1–G.4 present INT8 accuracy and F1 scores under increasing Gaussian noise severity in tabular form for both the original and expanded evaluation splits. The results support the trends described in Section 4.4: that FFT and LSKD remain closely aligned at low-to-moderate corruption levels, with LSKD achieving higher mean with overlapping variance at severe levels. BNLP follows a similar trajectory and achieves generally comparable performance to LSKD at severe levels, and LP shows comparatively higher instability at moderate severities but greater accuracy under extreme corruption levels. Results on the expanded evaluation set show the same overall patterns, confirming consistency across test-set size.

F1 scores follow the same general behaviour as accuracy, with slightly larger divergence at higher severities, reflecting uneven class-level degradation under stronger corruption.

Table G.1. INT8 test accuracy under Gaussian noise (original test set; mean $\pm$ standard deviation over five seeds).

Gaussian Noise	FFT	LSKD	BNLP	LP
0.02	91.11 $\pm$ 1.55	90.37 $\pm$ 1.45	87.16 $\pm$ 1.75	77.84 $\pm$ 1.52
0.03	81.32 $\pm$ 2.88	81.79 $\pm$ 1.96	77.74 $\pm$ 2.40	73.11 $\pm$ 1.64
0.04	68.26 $\pm$ 2.71	68.58 $\pm$ 2.33	65.89 $\pm$ 2.45	66.68 $\pm$ 2.83
0.05	52.26 $\pm$ 2.18	55.53 $\pm$ 3.66	53.53 $\pm$ 2.71	61.21 $\pm$ 1.44
0.06	40.26 $\pm$ 1.59	45.11 $\pm$ 2.38	45.58 $\pm$ 4.18	55.05 $\pm$ 1.45

Table G.2. INT8 F1 score under Gaussian noise (original test set; mean $\pm$ standard deviation over five seeds).

Gaussian Noise	FFT	LSKD	BNLP	LP
0.02	90.88 $\pm$ 1.61	90.08 $\pm$ 1.50	86.99 $\pm$ 1.74	77.76 $\pm$ 1.70
0.03	80.52 $\pm$ 2.90	80.96 $\pm$ 2.20	76.41 $\pm$ 2.83	72.53 $\pm$ 1.86
0.04	65.31 $\pm$ 2.71	66.05 $\pm$ 2.78	63.18 $\pm$ 2.39	65.96 $\pm$ 2.57
0.05	48.20 $\pm$ 2.68	52.26 $\pm$ 3.51	49.20 $\pm$ 3.05	59.59 $\pm$ 1.75
0.06	36.11 $\pm$ 3.21	40.87 $\pm$ 2.90	40.78 $\pm$ 3.98	53.54 $\pm$ 1.42

Table G.3. INT8 test accuracy under Gaussian noise (expanded split).

Gaussian Noise	FFT	LSKD	BNLP	LP
0.02	90.38 $\pm$ 1.39	89.56 $\pm$ 0.73	86.07 $\pm$ 1.60	76.34 $\pm$ 0.57
0.03	81.03 $\pm$ 2.69	81.18 $\pm$ 1.24	76.22 $\pm$ 1.85	71.80 $\pm$ 0.94
0.04	67.05 $\pm$ 2.35	68.24 $\pm$ 1.88	64.57 $\pm$ 1.58	65.25 $\pm$ 0.91
0.05	52.05 $\pm$ 2.33	54.71 $\pm$ 2.18	53.16 $\pm$ 1.93	58.99 $\pm$ 1.00
0.06	39.52 $\pm$ 1.36	42.75 $\pm$ 1.95	42.67 $\pm$ 1.64	52.19 $\pm$ 0.66

Table G.4. INT8 F1 score under Gaussian noise (expanded split).

Gaussian Noise	FFT	LSKD	BNLP	LP
0.02	90.17 $\pm$ 1.39	89.40 $\pm$ 0.68	85.92 $\pm$ 1.64	76.12 $\pm$ 0.58
0.03	80.00 $\pm$ 2.95	80.60 $\pm$ 1.40	75.23 $\pm$ 2.12	71.30 $\pm$ 1.10
0.04	64.57 $\pm$ 2.12	66.59 $\pm$ 2.09	62.38 $\pm$ 1.76	64.32 $\pm$ 0.87
0.05	48.73 $\pm$ 2.32	52.07 $\pm$ 2.29	50.09 $\pm$ 2.18	57.46 $\pm$ 1.02
0.06	35.63 $\pm$ 2.10	39.15 $\pm$ 2.05	38.68 $\pm$ 1.97	50.21 $\pm$ 1.01

G.2 JPEG Compression Results

Tables G.5 and G.6 present INT8 accuracy under increasing JPEG compression strength for the original and expanded evaluation splits. The tabular results confirm the trends reported in Section 4.4. Accuracy decreases progressively with stronger compression, with FFT and LSKD remaining closely aligned across most compression levels. BNLP shows moderate degradation at severe compression levels, while LP exhibits lower robustness overall. Results on the expanded evaluation set show generally similar behaviour, indicating consistent robustness patterns across test-set size.

Table G.5. INT8 test accuracy under JPEG compression (original split).

JPEG Compression	FFT	LSKD	BNLP	LP
50	89.95 $\pm$ 1.45	89.53 $\pm$ 1.43	85.84 $\pm$ 1.58	74.84 $\pm$ 0.88
40	88.11 $\pm$ 1.47	87.58 $\pm$ 0.97	84.21 $\pm$ 0.93	70.16 $\pm$ 1.81
30	85.68 $\pm$ 0.88	85.53 $\pm$ 1.10	79.68 $\pm$ 1.94	67.00 $\pm$ 1.39
20	76.53 $\pm$ 1.76	79.21 $\pm$ 2.49	70.68 $\pm$ 1.69	61.84 $\pm$ 0.85
10	53.63 $\pm$ 1.95	56.89 $\pm$ 3.08	44.11 $\pm$ 3.94	45.79 $\pm$ 0.67

Table G.6. INT8 test accuracy under JPEG compression (expanded split).

JPEG Compression	FFT	LSKD	BNLP	LP
50	89.07 $\pm$ 0.69	88.69 $\pm$ 1.11	85.14 $\pm$ 1.30	72.65 $\pm$ 1.12
40	87.28 $\pm$ 0.89	86.96 $\pm$ 1.44	82.91 $\pm$ 1.29	70.84 $\pm$ 0.93
30	84.07 $\pm$ 1.29	84.26 $\pm$ 1.60	78.11 $\pm$ 1.55	67.31 $\pm$ 1.13
20	75.81 $\pm$ 1.18	77.07 $\pm$ 2.54	69.43 $\pm$ 1.64	61.04 $\pm$ 1.06
10	54.50 $\pm$ 0.87	56.57 $\pm$ 3.19	45.67 $\pm$ 2.95	44.54 $\pm$ 0.84

Appendix H: Class-Level Results

This appendix examines whether transfer learning regimes exhibit different failure patterns at the class-level. Across regimes, the lowest-performing classes are largely shared, with several tomato disease categories consistently ranking among the most challenging. This suggests that class-level difficulty may be primarily data-intrinsic rather than regime-specific.

H.1 Class-level Results on the Original (Small) Test Set

Table H.1 reports representative low-performing classes on the 10-per-class test split. Due to the limited sample size, per-class rankings exhibit higher variance across seeds. Nevertheless, broadly similar qualitative patterns are observed: tomato disease categories frequently appear among the lowest-performing classes across FFT, BNLP, LSKD, and LP. Some regime-specific differences are present, but no regime demonstrates a fundamentally distinct failure profile.

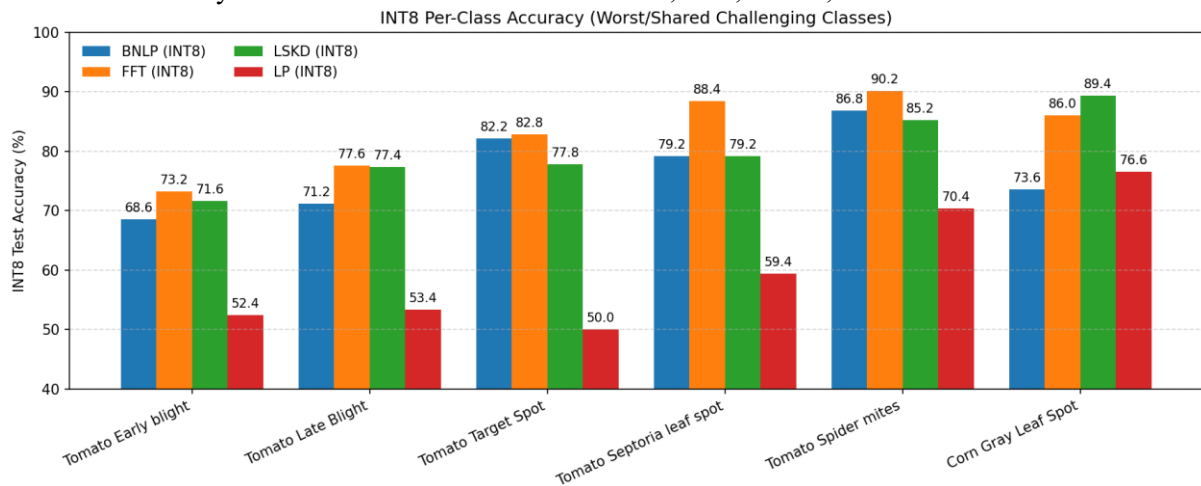
Table H.1. Selected low-performing classes (INT8 accuracy, mean $\pm$ standard deviation over five seeds; original test split).

Class	BNLP (INT8)	FFT (INT8)	LSKD (INT8)	LP (INT8)
Tomato Early blight	64.00 $\pm$ 11.40	82.00 $\pm$ 4.47	66.00 $\pm$ 16.73	46.00 $\pm$ 5.48
Tomato Septoria leaf spot	66.00 $\pm$ 11.40	82.00 $\pm$ 8.37	62.00 $\pm$ 13.04	64.00 $\pm$ 11.40
Corn Northern Leaf Blight	86.00 $\pm$ 8.94	84.00 $\pm$ 5.48	66.00 $\pm$ 5.48	62.00 $\pm$ 8.37
Peach Bacterial spot	78.00 $\pm$ 10.95	78.00 $\pm$ 4.47	76.00 $\pm$ 11.40	74.00 $\pm$ 5.48
Tomato Spider mites	76.00 $\pm$ 5.48	80.00 $\pm$ 7.07	84.00 $\pm$ 8.94	70.00 $\pm$ 0.00

H.2 Class-level Results on the Expanded Evaluation Set

Table H.2 reports overlapping low-performing classes on the expanded evaluation set (~100 images per class). The identity of the most challenging classes remains largely unchanged relative to the smaller split. Tomato-related disease categories consistently appear among the lowest-performing classes across regimes. While minor regime-specific variations and magnitude differences are observed in ranking, the dominant difficult classes are generally shared. These results indicate that class-level difficulties may not be substantially altered by the choice of transfer learning regime.

Figure H.1. INT8 per-class test accuracy for selected challenging classes on the expanded evaluation set. Mean accuracy across five seeds is shown for BNLP, FFT, LSKD, and LP.



For completeness, we report a table for common classes for the increased test set across the top-6 overlapping classes:

Table H.2. Common low-performing classes across regimes (INT8 accuracy, mean $\pm$ standard deviation over five seeds; expanded test split).

Class	BNLP (INT8)	FFT (INT8)	LSKD (INT8)	LP (INT8)
Tomato Early blight	68.60 $\pm$ 3.78	73.20 $\pm$ 3.63	71.60 $\pm$ 3.21	52.40 $\pm$ 2.88
Tomato Late Blight	71.20 $\pm$ 5.26	77.60 $\pm$ 6.91	77.40 $\pm$ 4.16	53.40 $\pm$ 7.64
Tomato Target Spot	82.20 $\pm$ 1.64	82.80 $\pm$ 2.77	77.80 $\pm$ 3.27	50.00 $\pm$ 4.85
Tomato Septoria leaf spot	79.20 $\pm$ 4.60	88.40 $\pm$ 4.34	79.20 $\pm$ 6.76	59.40 $\pm$ 6.15
Tomato Spider mites	86.80 $\pm$ 3.63	90.20 $\pm$ 3.96	85.20 $\pm$ 4.09	70.40 $\pm$ 3.05
Corn Gray Leaf Spot	73.60 $\pm$ 4.67	86.00 $\pm$ 4.00	89.40 $\pm$ 3.36	76.60 $\pm$ 2.70